

Project 1

AI-Powered Emotion Detection from Text

Project Overview

This project focuses on using **Natural Language Processing (NLP)** to detect the emotional tone of text. Students will train a model to classify text into categories such as *happy, sad, angry, surprised, neutral*, etc. This beginner-friendly project strengthens understanding of text preprocessing, NLP models, vectorization, and classification.

Problem Statement

Build an AI model that can read a sentence or paragraph and predict the underlying emotion expressed by the user.

Objectives

- Understand text preprocessing pipelines
 - Work with tokenization, cleaning, stopword removal
 - Build and train a classification model
 - Evaluate accuracy and interpret predictions
 - Visualize results with sample outputs
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Tech Stack

- Python
- NumPy, Pandas

- Scikit-Learn / TensorFlow
 - NLTK or SpaCy
 - Matplotlib/Seaborn for visualization
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◆ Workflow

1. Dataset Collection

Use open datasets (e.g., *Emotion Dataset*, *Twitter Sentiment Dataset*).

2. Text Preprocessing

- Lowercasing
- Removing special characters
- Tokenization
- Stopwords removal
- Stemming/Lemmatization

3. Feature Extraction

- Bag-of-Words
- TF-IDF
- Word embeddings (optional)

4. Model Building

- Naive Bayes / Logistic Regression
- OR a simple LSTM classifier

5. Model Evaluation

- Accuracy, F1 score, confusion matrix

- Example predictions

6. Final Output

A model that predicts emotion when a user inputs text.

◆ Deliverables

- Jupyter/Colab notebook
 - Data preprocessing scripts
 - Trained model + prediction samples
 - Final report explaining your approach
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◆ Real-World Applications

- Customer support sentiment analysis
- Social media emotion tracking
- Mental health early warning systems
- Chatbots and conversational AI